Exact Exchange in Levi–Perdew–Sahni Density Functional Theory

Ivan P. Bosko and Viktor N. Staroverov
Department of Chemistry, The University of Western Ontario, London, Ontario N6A 5B7, Canada
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In this work, we identified that the Fermi–Amaldi functional constitutes the exact orbital-free exchange energy within the Levy–Perdew–Sahni formalism for the square root of the electron density. Our analytical derivation relies on a specific choice of wave function describing a reference system of non-interacting boson-like fermions. This perspective preserves the conventional structure of Levy–Perdew–Sahni theory while rigorously defining the exact exchange term. This advancement enables the construction of orbital-free functionals containing exact exchange. Furthermore, our numerical study demonstrates that including the exact exchange in the Levy–Perdew–Sahni scheme improves the accuracy of self-consistent calculations.

I. INTRODUCTION

Kohn–Sham density functional theory (KS DFT) is exact in principle and widely adopted for its effective compromise between accuracy and expense [1–3]. However, the method hits a bottleneck for large systems: the computational cost scales nearly cubically with the electron number, making calculations on periodic structures or proteins prohibitively expensive [4]. The Levy–Perdew–Sahni (LPS) formulation provides a distinct advantage offering a pathway to near-linear scaling that remains compatible with standard KS codes [4–6]. It models the electronic system as N non-interacting bosons with the interacting energy functional given by

$$E[n] = T_{\mathrm{W}}[n] + \int v_{\mathrm{ext}}(\mathbf{r})n(\mathbf{r})d\mathbf{r} + J[n] + G[n], \quad (1)$$

where the kinetic energy of the non-interacting bosons given by the von Weizsäcker functional [7]

$$T_{\mathrm{W}}[n] = \int n^{1/2}(\mathbf{r}) \left(-\frac{1}{2}\nabla^2\right) n^{1/2}(\mathbf{r})\; d\mathbf{r} \quad (2)$$

and the classical Coulomb energy is given by

$$J[n] = \frac{1}{2}\int\int \frac{n(\mathbf{r})n(\mathbf{r}')}{|\mathbf{r}-\mathbf{r}'|}d\mathbf{r}d\mathbf{r}'. \quad (3)$$

The functional $G[n]$ reconciles the bosonic reference system with the true interacting electron density, and must be approximated. Minimizing the energy functional (1) with respect to the density $n(\mathbf{r})$ leads to a single equation for the square root of the electron density

$$\left\{-\frac{1}{2}\nabla^2 + v_{\mathrm{eff}}([n];\mathbf{r})\right\} n^{1/2}(\mathbf{r}) = \mu n^{1/2}(\mathbf{r}), \quad (4)$$

where μ is the negative of the ionization energy. The effective potential

$$v_{\mathrm{eff}}([n];\mathbf{r}) = v_{\mathrm{ext}}(\mathbf{r}) + \int \frac{n(\mathbf{r}')}{|\mathbf{r}-\mathbf{r}'|}d\mathbf{r}' + \frac{\delta G[n]}{\delta n(\mathbf{r})} \quad (5)$$

enforces the correct physical density [5]. Given the approximation to $G[n]$, Eq. (4) can be solved self-consistently in a manner analogous to the KS equations [2].

Although the functional $G[n]$ can be approximated as a whole, this is customary to decompose it into kinetic and exchange-correlation components with the last borrowed from KS DFT [8]

$$G[n] = T_{\mathrm{P}}[n] + E_{\mathrm{xc}}^{\mathrm{KS}}[n]. \quad (6)$$

The Pauli term, $T_{\mathrm{P}}[n]$, is defined as a difference between the exact non-interacting electronic kinetic energy functional, $T_{\mathrm{s}}[n]$ [3], and the von Weizsäcker term

$$T_{\mathrm{P}}[n] = T_{\mathrm{s}}[n] - T_{\mathrm{W}}[n]. \quad (7)$$

One of the popular choices for $G[n]$ is the Thomas–Fermi–Dirac-λ-Weizsäcker (TFDλW) approximation [9–12]

$$T_{\mathrm{P}}[n] = T_{\mathrm{TF}}[n] + (\lambda - 1)T_{\mathrm{W}}[n] \quad (8)$$

$$E_{\mathrm{xc}}^{\mathrm{KS}}[n] = E_{\mathrm{x}}^{\mathrm{LDA}}[n]. \quad (9)$$

In this model, $T_{\mathrm{P}}[n]$ is constructed as a linear combination of the TF kinetic energy functional [13, 14]

$$T_{\mathrm{TF}}[n] = \frac{3}{10}\left(3\pi^2\right)^{2/3}\int n^{5/3}(\mathbf{r})d\mathbf{r} \quad (10)$$

and the von Weizsäcker term, with a mixing parameter λ. $E_{\mathrm{xc}}[n]$ is approximated using the Dirac (LDA) exchange functional [15]

$$E_{\mathrm{x}}^{\mathrm{LDA}}[n] = \frac{3}{4}\left(\frac{3}{\pi}\right)^{1/3}\int n^{4/3}(\mathbf{r})d\mathbf{r}. \quad (11)$$

Alternatively, the Dirac exchange may be substituted by modern GGA or meta-GGA functionals, such as PBE [16, 17].

The widespread application of the LPS DFT necessitates accurate approximations for both the kinetic, $T_{\mathrm{P}}[n]$, and exchange-correlation, $E_{\mathrm{xc}}[n]$, components of the energy. Significant progress has been made for the kinetic part, ranging from local and semi-local to advanced machine-learned approximations [4, 6, 18]. While this is common to use the exchange-correlation functionals from KS DFT, their applicability to LPS DFT is limited by GGAs and meta-GGAs [8, 19]. The more accurate approximations, including highly-accurate hybrid functionals, require the exact exchange functional, which depends on the KS orbitals [3]. Thus, the orbital-free exact exchange is a highly desirable advancement for the LPS DFT framework.

II. DERIVATION

While the original LPS formalism was derived for bosons, the authors noted its potential generalization to other types of particles [5]. Here, we consider a fictitious system of non-interacting boson-like fermions with spin degeneracy $M = 2s + 1$ equal to the particle number N. This perspective preserves the conventional structure of the theory while yielding another exact term serving as the orbital-free exchange functional.

The non-interacting wave function for fermions with $M = N$ is a Slater determinant of N orthonormal spin-orbitals $\{\psi_i(\mathbf{x})\}$, defined as a product of a spatial function $\sqrt{n(\mathbf{r})/N}$ and a spin function $\{\sigma_i(\omega)\}$. Since there are N distinct spin states, the Pauli exclusion principle allows single spatial orbital for all particles. Consequently, the antisymmetry is confined entirely to the spin part of the wave function

$$\Psi(\mathbf{x}_1, \ldots, \mathbf{x}_N) = \frac{\Phi_{\text{HP}}(\mathbf{r}_1, \ldots, \mathbf{r}_N)}{\sqrt{N!}} \det\{\sigma_i(\omega_j)\}, \quad (12)$$

where the spatial part Φ_{HP} is the Hartree product [20].

Explicit choice of the wave function enables applying the KS exact exchange definition to the LPS framework

$$E_{\text{x}}^{\text{exact}}[n] = \langle \Psi_{\min} | \hat{V}_{\text{ee}} | \Psi_{\min} \rangle - J[n], \quad (13)$$

where $\hat{V}_{\text{ee}} = \sum_{i<j}^{N} |\mathbf{r}_i - \mathbf{r}_j|^{-1}$ is the two-electron operator, and $\Psi_{\min}$ is the energy-minimizing wave function. Since $\hat{V}_{\text{ee}}$ is spin-independent, we can apply it to our boson-like fermion wave function. Substituting Eq. (12) into Eq. (13) yields the Fermi–Amaldi functional [21]

$$E_{\text{x}}^{\text{FA}}[n] = -\frac{1}{2N} \int\int \frac{n(\mathbf{r})n(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} d\mathbf{r} d\mathbf{r}'. \quad (14)$$

The expression (14) is the sought-for exact orbital-free exchange functional. This result aligns with our previous findings [22, 23] that the FA functional is the exact exchange energy for particles of any spin when they share a single spatial orbital. The corresponding FA potential is given by

$$v_{\text{x}}^{\text{FA}}([n]; \mathbf{r}) = -\frac{1}{N} \int \frac{n(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} d\mathbf{r}, \quad (15)$$

where N is treated as a constant parameter [24–30].

The LPS energy functional of Eq. (1) can be fully recovered with the current choice of the wave function of Eq. (12). Kinetic, electron-nuclear, Coulomb self-repulsion, and the Fermi–Amaldi exchange terms are exact since they are the expectation values of the corresponding operators. The remaining term, $G[n]$, is an approximate functional needed to reconcile the boson-like fermion reference system with the true interacting electron density.

III. RESULTS AND DISCUSSION

A. Implementation Details

To implement the LPS scheme of Eq. (4), we adapted the open-source Psi4NumPy code [31, 32] by constructing the density matrix exclusively from the lowest-energy eigenfunction normalized to N [5]. The current work utilizes the spin-polarized formalism: the α and β density matrices are generated from the lowest-energy eigenfunctions normalized to N_α and N_β, respectively.

Subtracting $T_{\text{W}}[n]$ in the Eq. (7) enables solution of Eq. (4) by preserving the KS eigenvalue problem form even for kinetic energy functionals lacking an explicit Laplacian operator, such as the Thomas–Fermi (TF) functional [5, 33]. A Core Hamiltonian initial guess was employed, and all molecular integrals were computed analytically using a Gaussian-type basis set [20]. The approximate functionals $T_{\text{P}}[n]$ and $E_{\text{x}}[n]$ were evaluated numerically using a Lebedev $(999, 974)$ quadrature grid with a Treutler partition of the atomic weights, which is the default in Psi4NumPy. In difficult convergence cases, the electron density was calculated at smaller, $(350, 74)$, grid and was used as an initial guess for the larger grid calculations.

To obtain the self-consistent solutions, we used a combination of the direct inversion in the iterative subspace (DIIS) procedure [34] with Hartree damping [35] and Level-Shifting [36] techniques. The maximum length of the DIIS vector was varied from 3 to 9 depending on the system. All eigenvalues except the lowest one were shifted up by $0.1 - 0.9$ at the beginning of the SCF cycle returning it back to 0.0 closer to the convergence. The damping factor was varied in the range $0.1 - 0.999$. The convergence criteria for both the total energy and RMSD of the density matrix was set to 1.0E-5 Hartree and unitless change, respectively.

To validate our implementation, we reproduced the reference H–Ne results of Chan *et al.* [10] and Karasiev *et al.* [9]. For open-shell atoms, this required adapting our code to match the spin-restricted approach used in those studies.

B. Self-Consistent Atomic Energies

Table I contains the mean absolute errors MAE $= \left(\sum_a \Delta E_a\right)/N_A$, where $\Delta E_a = |E_a^{\text{LPS}} - E_a^{\text{UHF}}|$, across $N_A = 10$ atoms, in a.u., and the relative mean absolute errors rMAE $= \left(-\sum_a \Delta E_a / E_a^{\text{UHF}}\right)/N_A \times 100\%$ expressed as a percentage of the UHF/UGBS energy. To maintain the spherical symmetry of the atoms and assist the convergence, an ad hoc basis set consisting of 31 s-type Gaussians, $\exp(-\alpha r^2)$ ($\alpha \in [1.14\times10^7, 2.00\times10^{-2}]$), taken from the UGBS basis set [37–40] was used.

Specific values of λ were adopted from the literature: $\lambda = 1$, the original Weizsäcker value [7], known to substantially overestimate the kinetic energy [18, 41];

TABLE I. H–Ne atomic SCF energy errors (relative to UHF/UGBS) for Fermi–Amaldi, LDA, and PBE exchange functionals combined with TFλW and Tran–Wesołowski (TW) kinetic energy functionals using the basis set of 31 s-type basis functions for all atoms (see Sec. III B).

λ	1/9	1/6	0.186	0.2	1/3	1.0	TW
			Mean Absolute Error (a.u.)				
LDA	4.38	1.58	0.76	0.24	4.23	15.74	6.61
PBE	4.95	2.11	1.26	0.69	3.78	15.44	7.16
FA	2.91	0.23	0.65	1.19	5.48	16.67	5.17
			Relative Mean Absolute Error (%)				
LDA	9.67	3.72	2.09	1.72	12.24	40.24	16.28
PBE	12.54	5.10	3.03	1.88	10.07	38.86	19.03
FA	8.00	1.14	1.66	2.71	13.41	41.00	14.57

$\lambda = 1/3$, derived from a two-point trapezoidal discretization of the density matrix [12]; $\lambda = 1/5$, determined empirically for hydrogenic atoms [11, 42]; $\lambda = 0.186$, corresponding to the large-Z asymptotic limit [43]; $\lambda = 1/9$, the coefficient for the second-order gradient expansion [3]; and $\lambda = 1/6$, which approximates the inclusion of the fourth-order correction [44]. Despite known deficiencies [4], the TFλW model was selected because its self-consistent convergence remains manageable within modified KS codes [9, 10].

In alignment with other authors [9, 10], Table I shows that the optimal von Weizsäcker mixing parameter for the PBE and LDA pure exchange functionals is $\lambda = 1/5$. As for the FA functional, the optimal value is $\lambda = 1/6$, with the second best being $\lambda = 0.186$. This result indicates that the FA functional can outperform both LDA and PBE exchange functionals when paired with an appropriate kinetic energy density functional.

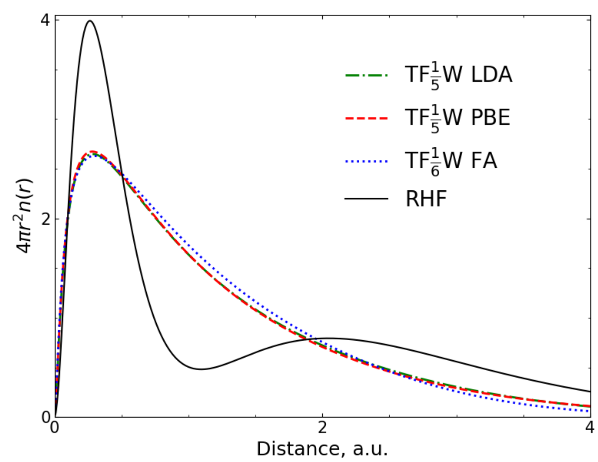

FIG. 1. Be SCF radial distribution function $4\pi r^2 n(\mathbf{r})$ for Fermi–Amaldi, LDA, and PBE exchange functionals combined with TF$\frac{1}{6}$W and TF$\frac{1}{5}$W kinetic energy functionals, respectively. The basis set with 31 s-type basis functions was used (see Sec. III B). The RHF reference is included for comparison.

C. Atomic Densities

Self-consistent radial distribution functions for the Be atom are presented in Fig. 1. The optimal TFλW kinetic functionals were employed for each exchange functional. All calculations used the basis set with 31 s-type basis functions described in the previous section.

The TF$\frac{1}{6}$W FA density decays more rapidly than either the TF$\frac{1}{5}$W LDA or TF$\frac{1}{5}$W PBE densities. As expected, the choice of exchange functional does not fundamentally alter the density structure, i.e., the FA exchange does not recover atomic shell structure, which remains absent due to the monotonic nature of the TFλW kinetic model. The more rapid asymptotic decay observed in the FA density is consistent with established results from statistical atomic theory [45].

D. Dissociation Profiles

Figure II shows the H_2 dissociation curves computed using the same models as in the previous section for density plotting. These calculations employed the basis set containing 13 s- and 4 p-functions taken from the UGBS basis set ($\alpha_s \in [9.34 \times 10^2, 7.67 \times 10^{-2}]$ and $\alpha_p \in [5.76 \times 10^{-1}, 3.92 \times 10^{-2}]$) for each hydrogen. An RHF curve with the same basis set is included as the reference. Although spin-polarized functionals were used, our LPS implementation lacks a symmetry-breaking mechanism. As such, it converges to the spin-symmetric solution, making RHF the appropriate spin-symmetric reference rather than the broken-symmetry UHF solution.

TABLE II. H–Ne atomic post-SCF energy errors (relative to UHF/UGBS) for Fermi–Amaldi, LDA, and PBE exchange functionals combined with TFλW and Tran–Wesołowski (TW) kinetic energy functionals. SCF densities were obtained at the UHF/UGBS level of theory.

λ	1/9	1/6	0.186	0.2	1/3	1.0	TW
			Mean Absolute Error (a.u.)				
LDA	0.37	2.29	2.95	3.44	8.04	31.05	0.52
PBE	0.21	1.78	2.45	2.93	7.53	30.54	0.07
FA	1.91	3.83	4.50	4.98	9.58	32.59	2.06
			Relative Mean Absolute Error (%)				
LDA	3.02	7.86	9.54	10.76	22.36	80.37	2.97
PBE	0.93	5.41	7.09	8.31	19.91	77.92	0.60
FA	4.29	9.12	10.80	12.02	23.62	81.63	4.24

As shown in Figure II, the FA functional exhibits better long-range behavior and a distinct potential well unlike both the LDA and PBE functionals. The FA functional also yields an equilibrium distance, r_e, significantly closer to the RHF reference, e.g., 2.00 a.u. against 3.00 and 3.25 a.u. for LDA and PBE, respectively. The FA binding energy estimate, $E(4\,\text{a.u.}) - E(r_e) = 0.0134$ a.u., is closer to the RHF reference (0.2191 a.u.) compared to 0.0028 and 0.0007 a.u. for LDA and PBE, respectively.

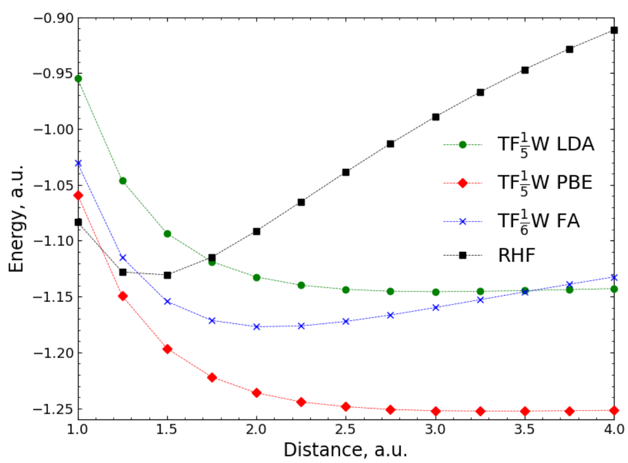

FIG. 2. H_2 SCF potential energy surface for Fermi–Amaldi, LDA, and PBE exchange functionals combined with TF$\frac{1}{6}$W and TF$\frac{1}{5}$W kinetic energy functionals, respectively. Calculations were performed with the basis set containing 13 s- and 4 p-functions (see Sec. III D). Self-consistent calculations were performed only at the marked points. The connecting curves are interpolations.

E. Post-SCF Atomic Energies

The FA functional's promising energetic performance on one hand with the absence of shell structure in the self-consistent densities on another prompted us to evaluate its accuracy in post-SCF calculations. The set of self-consistent UHF/UGBS atomic densities for H–Ne was used to compute post-SCF atomic energies with the TFλW kinetic energy functional and thex FA, LDA, and PBE exchange functionals.

Table II summarizes the mean absolute errors (MAE) and relative mean absolute errors (rMAE) for each exchange-kinetic combination. Overall, the performance of the approximate exchange functionals strongly depends on the choice of the kinetic functional. The FA functional yields consistently higher errors than both LDA and PBE for all tested kinetic functionals. When paired with the TFλW model, all exchange functionals achieve their lowest MAE and rMAE at $\lambda = 1/9$ which corresponds to the second-order gradient expansion of the TF functional.

This result does not contradict the self-consistent findings of the previous section, where the self-consistent densities exhibited no shell structure. The post-SCF calculations are performed with the accurate fermionic densties, for which the FA functional is not exact and known to be less accurate than LDA and PBE [46].

IV. CONCLUSION

In this work, we have reinterpreted the Levy–Perdew–Sahni (LPS) scheme to demonstrate that the Fermi–Amaldi functional is the orbital-free exact exchange functional within the LPS framework. By utilizing a reference system of non-interacting boson-like fermions, where all particles share a single spatial orbital, we showed that the conventional LPS energy functional [5] is fully recovered and another exact functional is easily defined.

The Fermi–Amaldi functional exhibits several highly desirable properties for an exchange functional. It is self-interaction free for one-orbital systems [22, 46], possesses the correct long-range asymptotic behavior [46], and exhibits the necessary derivative discontinuity at integer electron numbers N [24]. Furthermore, unlike the exact Kohn–Sham exchange potential, the Fermi–Amaldi potential is strictly multiplicative. Recently, we also extended the FA functional to fractional electron number [24]. In this work we also showed that the FA functional can be more accurate than LDA and PBE in self-consistent calculations. Collectively, these features identify the FA functional as a promising building block for future developments in LPS DFT.

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